

Clinical Pattern Recognition and Decision Support System

Healthcare fraud represents a major global financial burden, accounting for an estimated 9% to 19% of total healthcare expenditures in the United States alone (Hamid et al., 2024). The complexity of fraud schemes, ranging from identity theft to collusion and billing for unperformed services, poses significant challenges for detection. Analyzing large-scale healthcare data is further complicated by issues such as poor data quality, limited real-time detection capabilities, and inconsistent coding standards. A persistent obstacle in the field is the scarcity of labeled datasets, which limits the application of supervised learning techniques. Recent studies have leveraged unsupervised approaches; for example, one study utilized the CMS 2008-2010 DE-SynPUF dataset, focusing on inpatient claims, which are known to be both more prevalent and costly than outpatient fraud (Hoseini et al., 2020; Hoseini et al., 2024; Shrestha et al., 2020). The authors selected key features, including provider and physician information, payment amounts, and diagnostic codes (Hamid et al., 2024). Their two-stage methodology first applied association rule mining (the Apriori algorithm) to extract frequent interaction patterns among patients, providers, and physicians, then used four unsupervised classifiers, such as Isolation Forest, Cluster-Based Local Outlier Factor, ECOD, and One-Class SVM, to identify anomalies indicative of fraudulent activity. This approach proved more efficient than baseline anomaly detection, with reduced processing time, and uniquely integrated interactions across all stakeholders in the healthcare "insurance triangle. However, limitations remain: the dataset's diversity may not fully represent the range of real-world fraud scenarios, potentially leading to biased models, and unsupervised methods still require human experts to validate suspected fraud (Hamid et al., 2024).

Healthcare fraud in billing and claims remains a persistent and costly issue for the U.S. healthcare system, with estimated annual losses reaching \$68 billion, approximately 3–10% of total healthcare expenditure (TASHIN et al., 2024). Common fraudulent practices include upcoding, phantom billing, duplicate claims, and unbundling of services. Beyond the direct financial toll, such fraud diverts critical resources from patient care, raises insurance premiums, and erodes public trust. Traditional detection approaches, such as manual audits and rule-based systems, are labor-intensive, struggle to scale with the high volume of claims, and lack the flexibility to adapt to evolving fraud strategies. In response, recent research has taken a data-driven approach, utilizing a wide array of structured claims data (billing codes, provider and patient information, reimbursement amounts), unstructured data (clinical notes and EHRs processed with NLP), historical claims patterns for risk scoring, and synthetic data to overcome privacy limitations and simulate complex fraud scenarios. The authors propose a multi-layered defense strategy that integrates advanced technological, operational, and policy measures, shifting from reactive to proactive fraud prevention. AI-powered tools and machine learning models, spanning supervised, unsupervised, and semi-supervised approaches, are leveraged for real-time anomaly detection and to identify patterns that traditional systems may miss. Operational strategies include provider risk scoring and automated claims triage to prioritize high-risk cases, thereby reducing false positives and improving resource allocation. Real-time monitoring and NLP integration further enhance accuracy and reduce manual errors. Despite these innovations, challenges persist: high false-positive rates still require human review, data privacy regulations (e.g., HIPAA, GDPR) limit data sharing, and technical barriers such as algorithmic bias, model opacity, and the need for continuous model updates remain. Furthermore, regulatory frameworks often lag behind technological advances, particularly in rapidly evolving domains like telehealth and AI-driven fraud detection (TASHIN et al., 2024).

Medicare and Medicaid fraud pose a significant threat to the U.S. healthcare system, resulting in annual losses exceeding \$60 billion and undermining the integrity of public health programs (Pingili, 2025). Traditional detection methods, such as rule-based systems and manual audits, are increasingly inadequate due to the scale, administrative complexity, and evolving nature of fraudulent billing practices. To address these vulnerabilities, recent studies have leveraged real-world Medicaid claims data from outpatient facilities, integrating diverse data sources, including claims records, patient demographics, provider details, and external datasets such as fraud alerts and regulatory guidelines. Key analytical features included Current Procedural Terminology (CPT) codes, Diagnosis-Related Groups (DRGs), and billing frequencies, with robust anonymization protocols in place. The author developed an AI-driven claims intelligence framework using unsupervised machine learning, specifically clustering algorithms and autoencoders, to autonomously identify anomalies in billing patterns and CPT code usage without relying on labeled data (Pingili, 2025). This approach achieved 91% accuracy and improved fraud detection by 52% compared to manual processes, while reducing manual audit workloads by 40% and yielding an estimated 3:1 return on investment. However, interpretability remains a challenge, as the "black box" nature of unsupervised models can undermine auditors' trust. Additional limitations include inconsistent data standards and terminologies, potential algorithmic bias leading to unfair outcomes, organizational and technical barriers to integrating AI with legacy systems, and the ongoing need for human oversight to validate and refine model outputs (Pingili, 2025).

In the United States, unbundling and upcoding result in billions of dollars in annual losses for health insurance companies, while the complexity and sheer volume of medical billing compliance policies make real-time manual auditing by human data analysts impractical (Machireddy et al., 2022; Rahaman et al., 2025; Adams et al., 2002). To address the challenges of fraud detection and compliance in federal CMS outpatient data, I propose a robust, multi-layered solution that leverages LangGraph, Llama 3.3 (70B), ChromaDB, FastAPI, Groq LLM, SQLite execution, and PII guardrails on the CMS DE-SynPUF (Data Entrepreneurs' Synthetic Public Use File) outpatient dataset. The process begins with an advanced ingestion pipeline that efficiently processes wide-format federal CMS datasets. This engine trims character padding inconsistencies and normalizes crucial parameters, such as Claim ID, Patient ID, Service Date, and Payment Amount, into optimized local staging tables. In the processing phase, the system goes beyond isolated claims analysis by deploying a Choreographer Timeline Pattern that dynamically clusters claims by composite keys (patient_id, service_date) into precise, chronological patient-day clinical windows. This approach transforms a massive dataset into highly manageable memory partitions, enabling deeper context-aware analysis. For auditing, the solution employs a Dual-Core Compliance Matrix. The Deterministic Gate leverages an in-memory registry of over 4,300 CMS NCCI edits, using Python set intersections to instantly flag unauthorized procedural splits, while the Probabilistic Gate profiles provider billing behavior against national benchmarks using Gaussian distribution analysis to identify outliers with a Z-score exceeding 2.84. These analytical streams are mapped onto a secure streamlit-based web interface, integrated with an independent AI microservice-based chatbot. The dashboard provides real-time telemetry (e.g., active flags, blocked overpayments (financial risk blocked), number of NCCI PTP violations (Unbundling), Upcoding, and interactive review queues), while an embedded LangGraph agent automates text-to-SQL translation, seamlessly transforming auditor queries into actionable insights with a latency of some seconds. Overall, this web-based solution streamlines data handling and compliance monitoring and empowers auditors with intuitive, AI-driven investigative tools for enhanced fraud detection and operational efficiency.

Bibliography

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Appendix

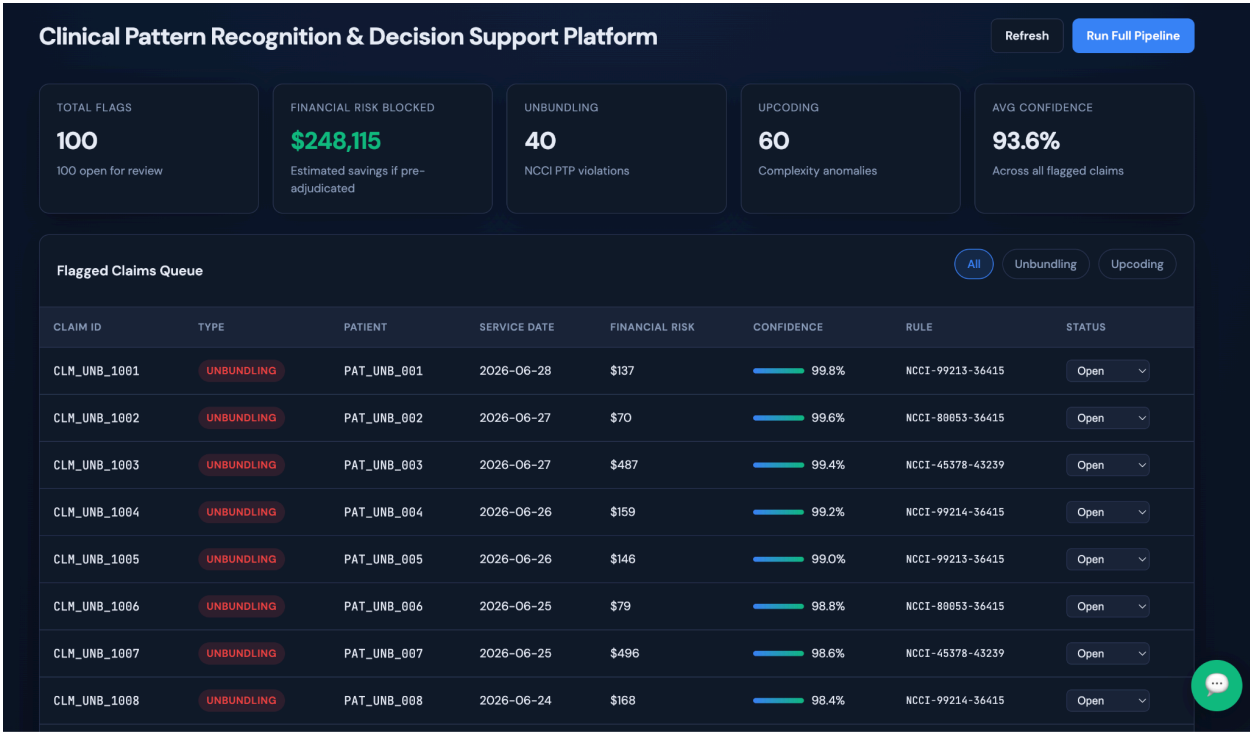


Figure 1. Clinical Pattern Recognition and Decision Support System and List of Flagged Claims

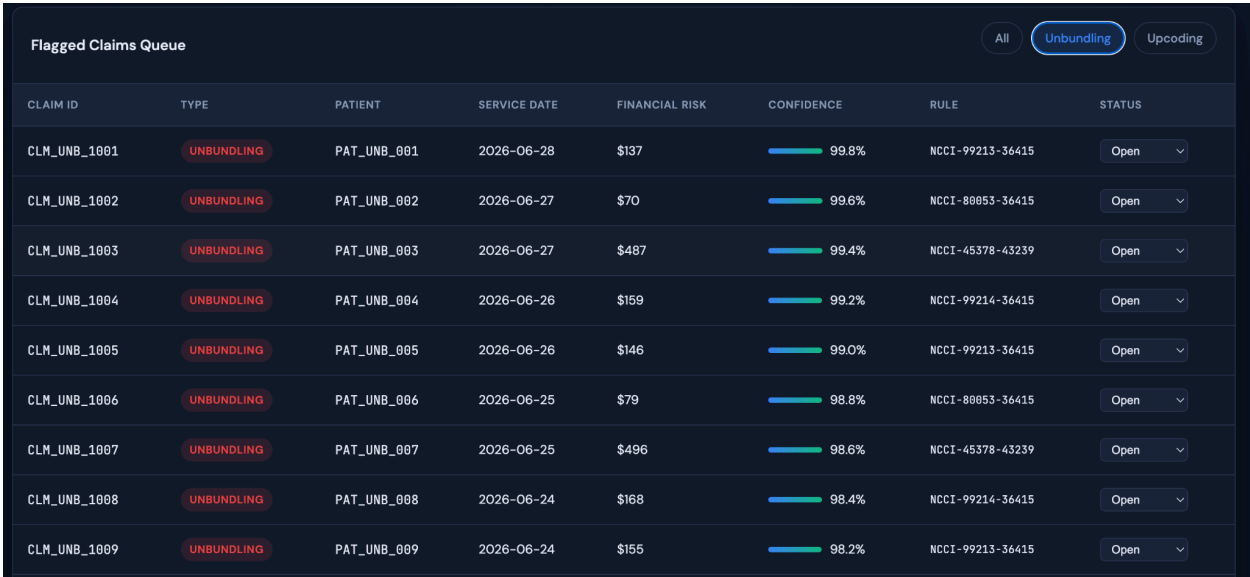


Figure 2. Flagged Claims listing: Unbundling

CLAIM ID	TYPE	PATIENT	SERVICE DATE	FINANCIAL RISK	CONFIDENCE	RULE	STATUS
CLM_UPC_2881	UPCODING	PAT_UPC_001	2026-06-28	\$3,669	89.2%	UPCODE-E&M	Open
CLM_UPC_2882	UPCODING	PAT_UPC_002	2026-06-28	\$3,684	89.1%	UPCODE-E&M	Open
CLM_UPC_2883	UPCODING	PAT_UPC_003	2026-06-27	\$3,698	89.0%	UPCODE-E&M	Open
CLM_UPC_2884	UPCODING	PAT_UPC_004	2026-06-27	\$3,713	88.9%	UPCODE-E&M	Open
CLM_UPC_2885	UPCODING	PAT_UPC_005	2026-06-27	\$3,728	88.8%	UPCODE-E&M	Open
CLM_UPC_2886	UPCODING	PAT_UPC_006	2026-06-26	\$3,743	88.7%	UPCODE-E&M	Open
CLM_UPC_2887	UPCODING	PAT_UPC_007	2026-06-26	\$3,757	88.6%	UPCODE-E&M	Open
CLM_UPC_2888	UPCODING	PAT_UPC_008	2026-06-26	\$3,772	88.5%	UPCODE-E&M	Open

Figure 3. Flagged Claims listing: Upcoding

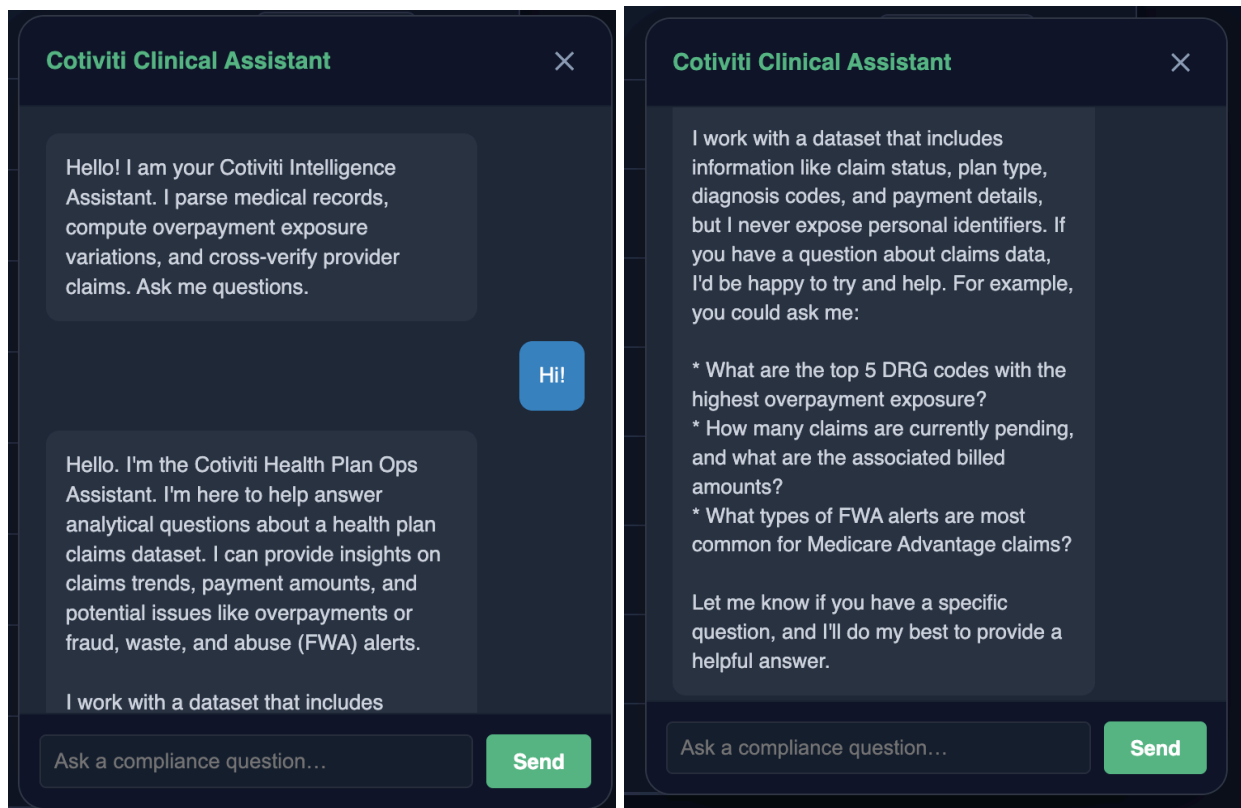


Figure 4. Chat Interface

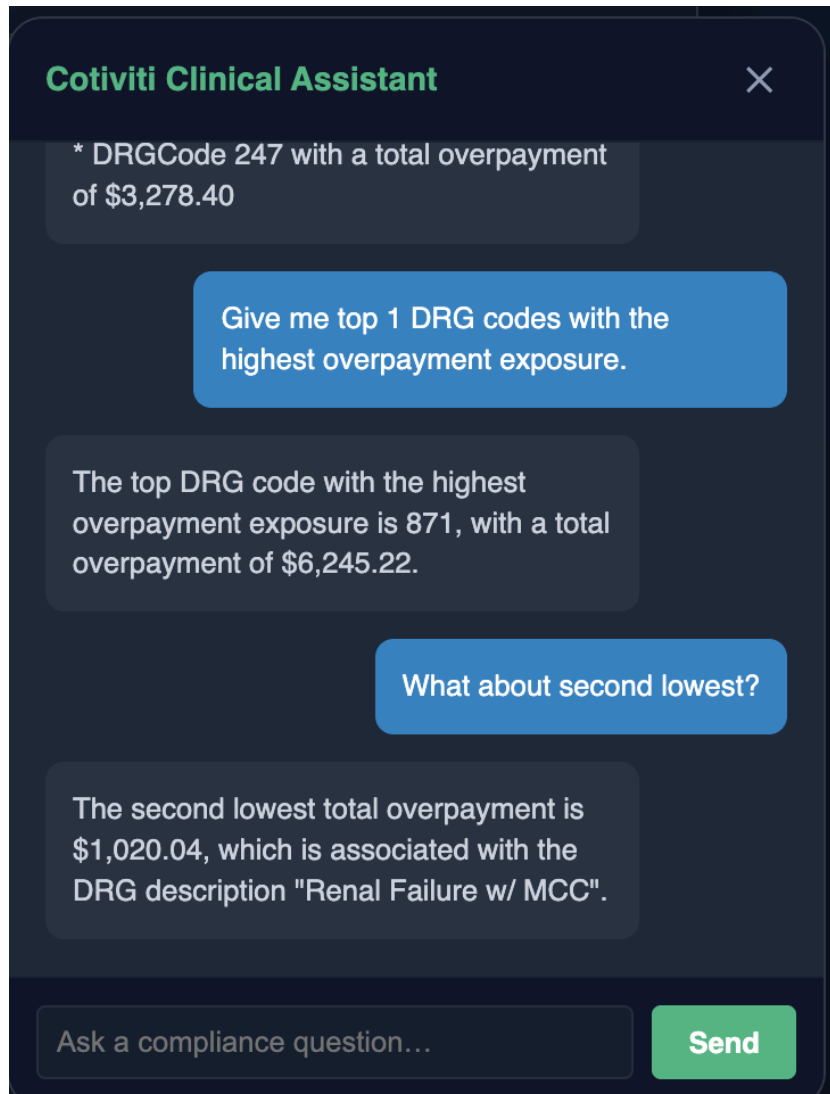


Figure 5. Questions and Answers: Chat Interface

Generated SQL for Validation

Interpreted Context Pattern: *What is the second lowest DRG code based on total overpayment amount?*

Extracted Context Token Signatures: DRG code , total overpayment amount , second lowest

Target Columns Retrieved from ChromaDB Index: OverpaymentAmt , DRGDescription , OverpaymentFlag , DRGCode , BilledAmount , PaidAmount , AllowedAmount , PrimaryDx

```
SELECT "DRGCode", SUM("OverpaymentAmt") AS "TotalOverpayment"
FROM claims
GROUP BY "DRGCode"
ORDER BY "TotalOverpayment" ASC
LIMIT 1 OFFSET 1
```

Refinement Optimization Loop Counts: 1

Total Query Pipeline Latency: 2.04s

Figure 6. SQL Generation and Latency: Chat Interface

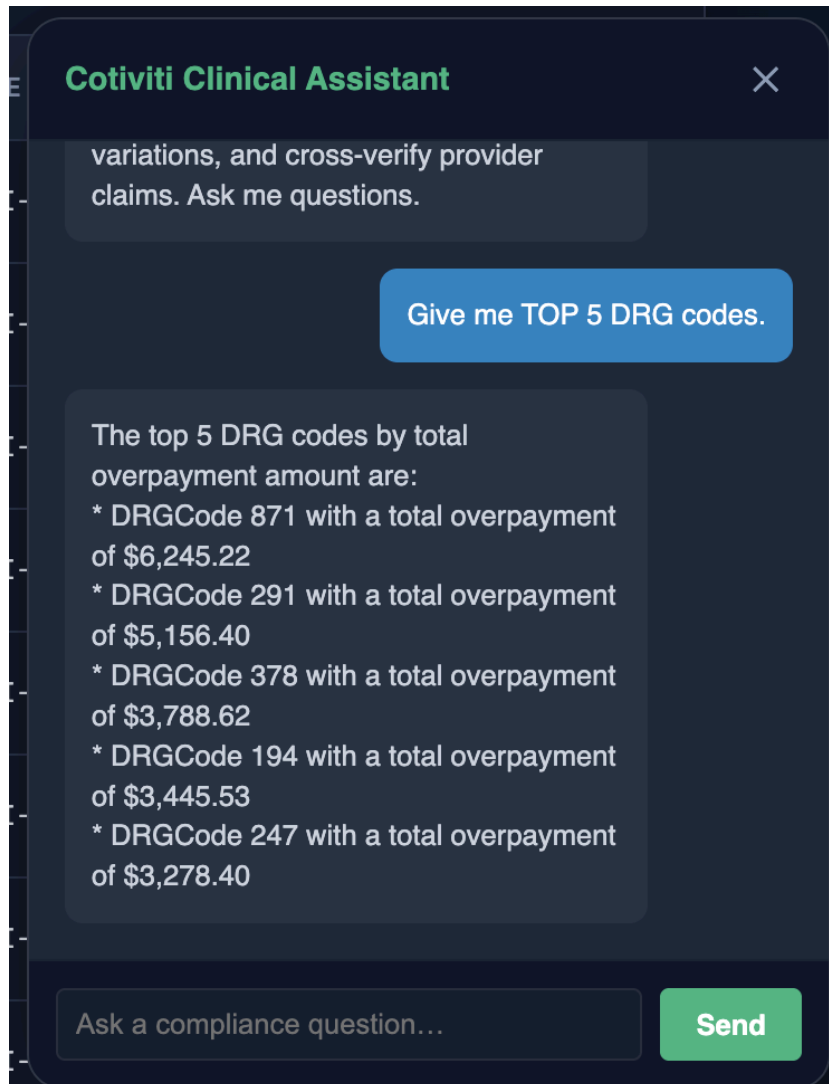


Figure 7. Questions and Answers: Chat Interface

Generated SQL for Validation

Extracted Context Token Signatures: TOP , 5 , DRG codes

Target Columns Retrieved from ChromaDB Index: DRGDescription , DRGCode , PrimaryDx , FWAType , PlanType , BilledAmount , ProviderNPI , OverpaymentAmt

```
SELECT "DRGCode", SUM("OverpaymentAmt") AS "TotalOverpayment"
FROM claims
GROUP BY "DRGCode"
ORDER BY "TotalOverpayment" DESC
LIMIT 5
```

Refinement Optimization Loop Counts: 1

Total Query Pipeline Latency: 2.05s

Query claim fields, active overpayment instances, or anomalous DRG codes...

↑

🧹 Clear Conversation Memory Workspace

Figure 8. SQL Generation and Latency: Chat Interface